



Lora Enabled Oscilloscope Smart Wearable

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Introduction

This project presents the design of a multi-functional, wearable diagnostic tool integrated into a compact wrist-worn form factor. Built on the ESP32-CAM platform, the device functions as a portable digital oscilloscope while integrating LIDAR-based ranging, 6-axis motion tracking, and LoRa-based long-range communication. It serves as an "Engineer's Swiss Army Knife," allowing for real-time signal analysis and environmental sensing in the field.

Objectives

- To develop a wearable oscilloscope capable of signal acquisition and OLED visualization.
- To integrate Long Range (LoRa) technology for wireless data transmission in remote areas.
- To implement precise distance measurement using the VL53L1X Time-of-Flight (ToF) sensor.
- To monitor orientation and motion dynamics using the MPU6050 IMU.
- To optimize power and pin management on the GPIO-limited ESP32-CAM.

Methodology

- Signal Conditioning:** Utilizing the LM358 Op-Amp as a unity-gain buffer to protect the ESP32 ADC and stabilize input signals.
- Data Acquisition:** Using high-speed ADC sampling on GPIO pins to capture analog waveforms.
- Communication Hub:** Configuring a shared I2C bus for the 0.96" OLED, MPU6050, and VL53L1X, while dedicating a custom SPI bus for the SX1278 LoRa module.
- Software Logic:** Developing a menu-based UI to toggle between Oscilloscope, Rangefinder, and Motion Tracking modes.

Societal Impact

- Education:** Provides an affordable, portable lab tool for students in rural or low-resource areas.
- Disaster Management:** LoRa connectivity allows for data transmission during cellular network failures.
- Industrial Safety:** Wearable sensors can detect falls (MPU6050) or proximity hazards (VL53L1X) for factory workers.
- Field Engineering:** Enables technicians to debug circuits in the field without carrying heavy equipment.

Functionality Flowchart: LoRa Enabled Oscilloscope Smart Wearable

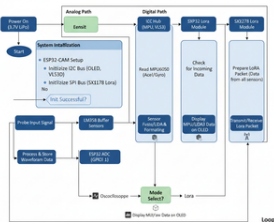


Fig1 FUNCTIONALITY FLOWCHART

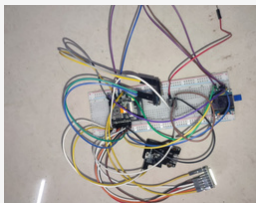


Fig 2 PROTOTYPE

Conclusions

The project successfully demonstrates that a high-utility diagnostic tool can be miniaturized into a wearable format. By overcoming the GPIO constraints of the ESP32-CAM and utilizing the LM358 for signal integrity, we created a stable platform that balances real-time visualization with long-range data telemetry.

Future Work

- PCB Fabrication:** Transitioning from a breadboard to a multi-layer flexible PCB for a true "watch" form factor.
- Advanced UI:** Implementing a touch-capacitive screen for better oscilloscope scale control (V/div and Time/div).
- Cloud Integration:** Using a LoRaWAN gateway to push field data to a centralized IoT dashboard.
- Extended Battery Life:** Implementing "Deep Sleep" modes and specialized power management ICs.